The Topology & Geometry of Autoencoders

Jacob Skarby

September 25, 2026

Munich Center for Machine Learning

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- Third year PhD student in TDA and machine learning
- One of four MCML junior member representatives
- Another interest: geopolitics and climate policy
- Climbing enthusiast

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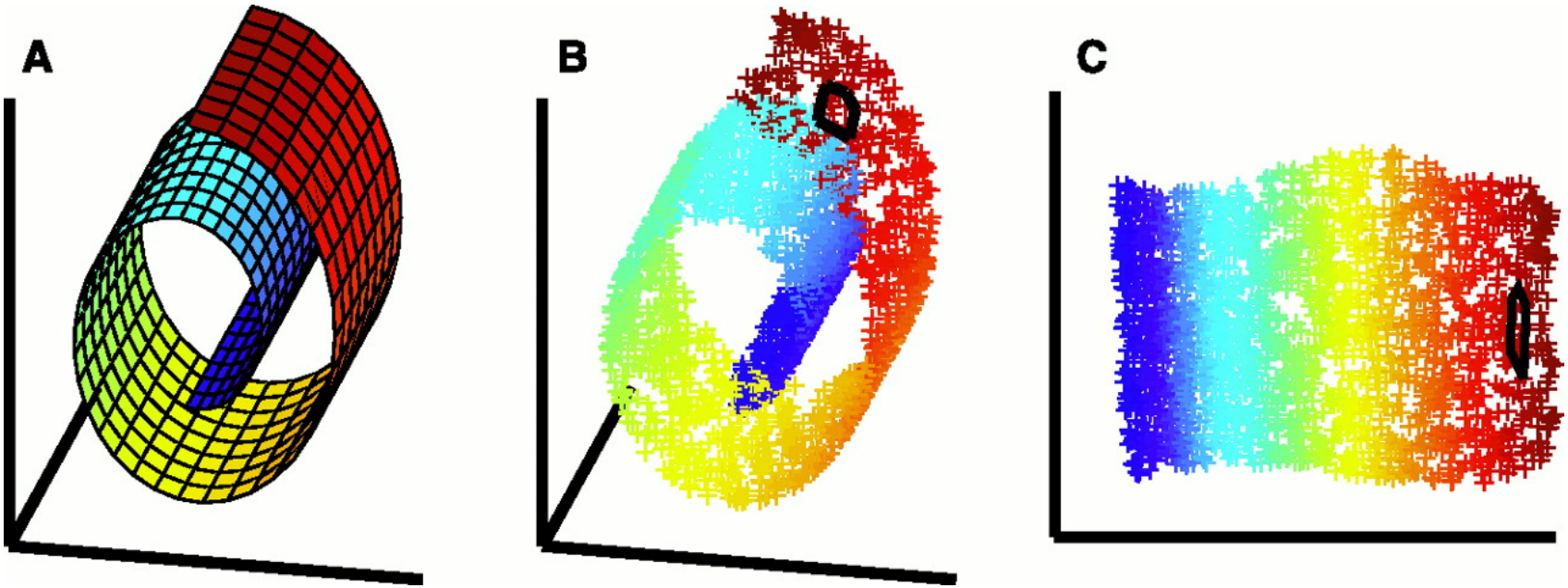
- Input: data set $X \in \mathbb{R}^N$ for some (possibly very large) N.
- Output: latent representation in $\mathbb{R}^d$ ($d < N$).



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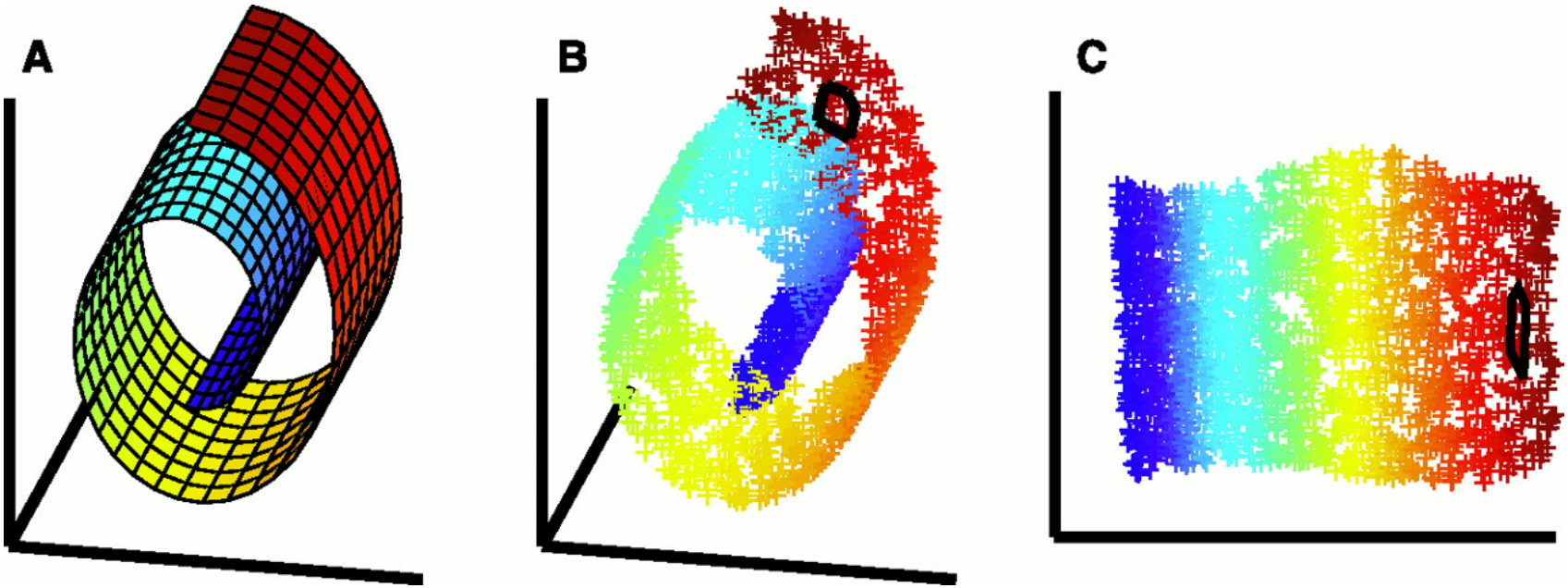
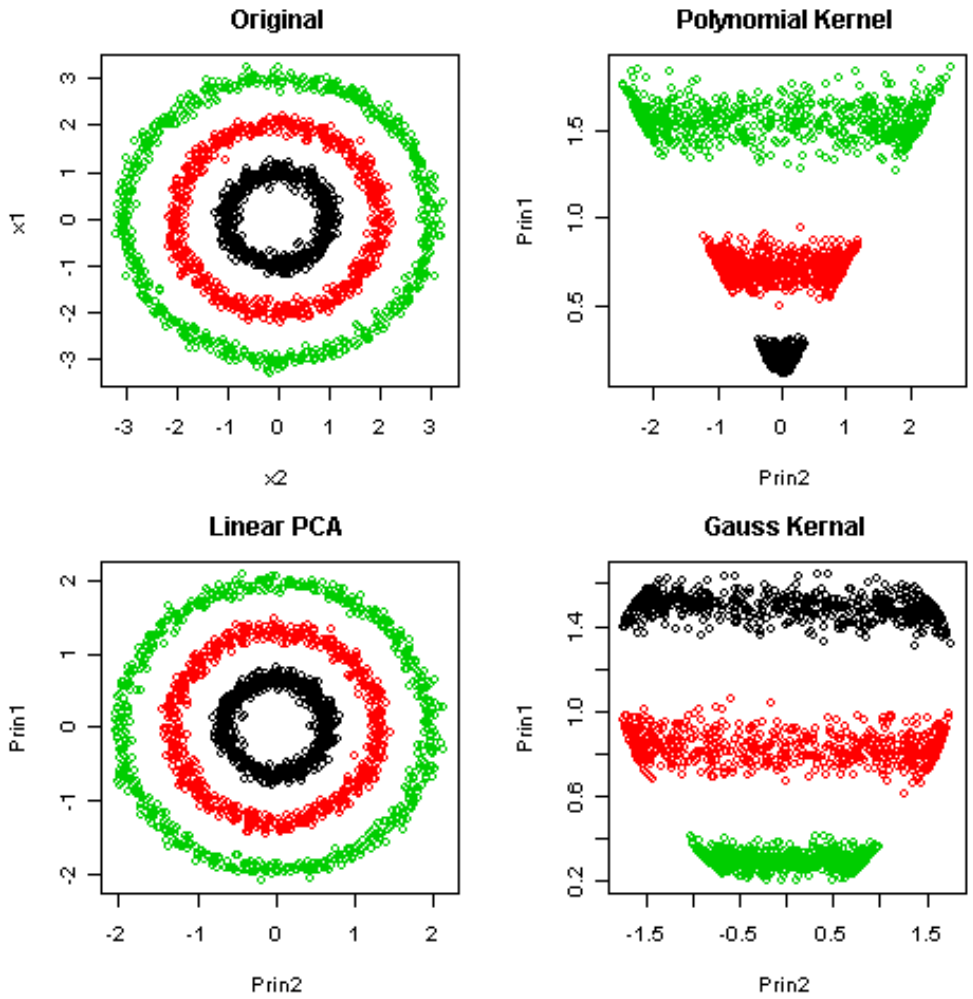
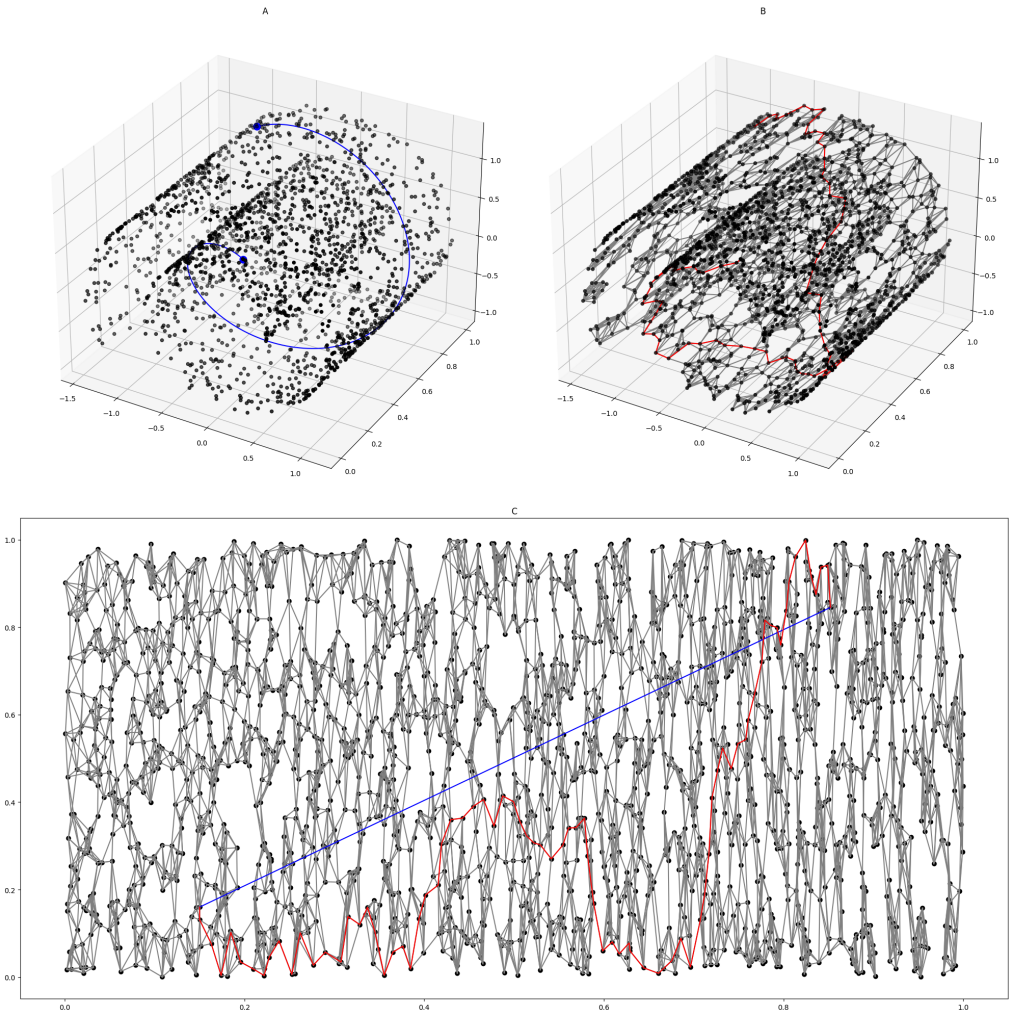


- Kernel PCA - transform (using some kernel), then do PCA



Popular NLDR methods

- Isomap [TdSL00] - essentially metric MDS using the geodesic distances



Popular NLDR methods

Other popular methods...

- t-SNE [vdMH08] (enjoyed by life scientists)
- UMAP [MHSG18] (based on a *category theoretic* framework)
- PHATE [MvDW+19] (useful for biological data)
- And so on...

Common theme: *just* a map into a latent space, no reconstruction/generative model.

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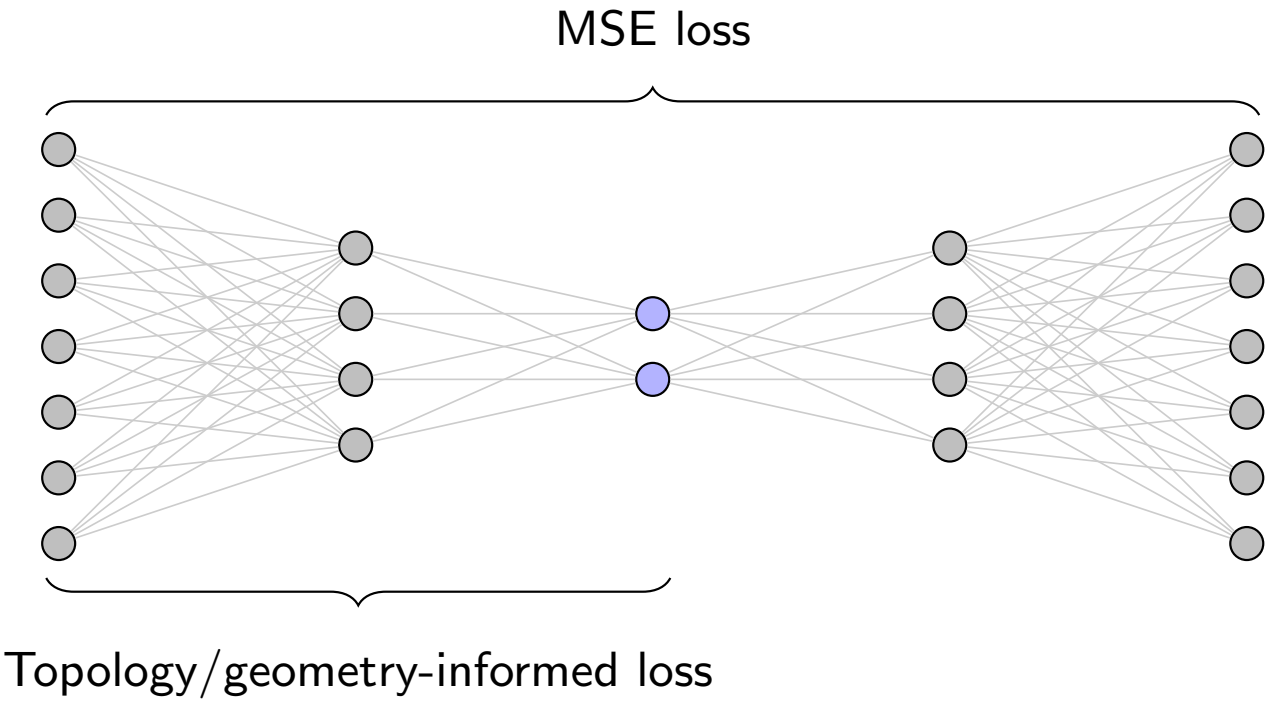
Key observation: *undercomplete autoencoders*

$$\mathbb{R}^N \xrightarrow{E} \mathbb{R}^d \xrightarrow{D} \mathbb{R}^N$$

perform NLDR. *Modify loss function* to make it do what we want!



Autoencoder-based NLDR

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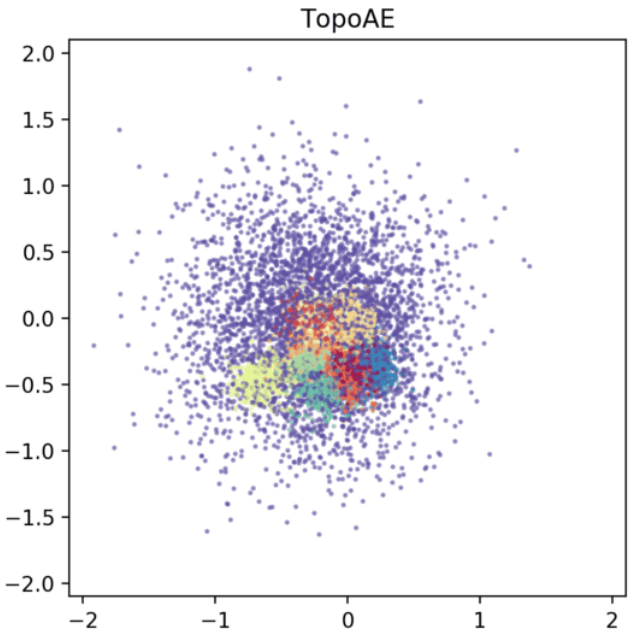
Plenty of existing approaches!



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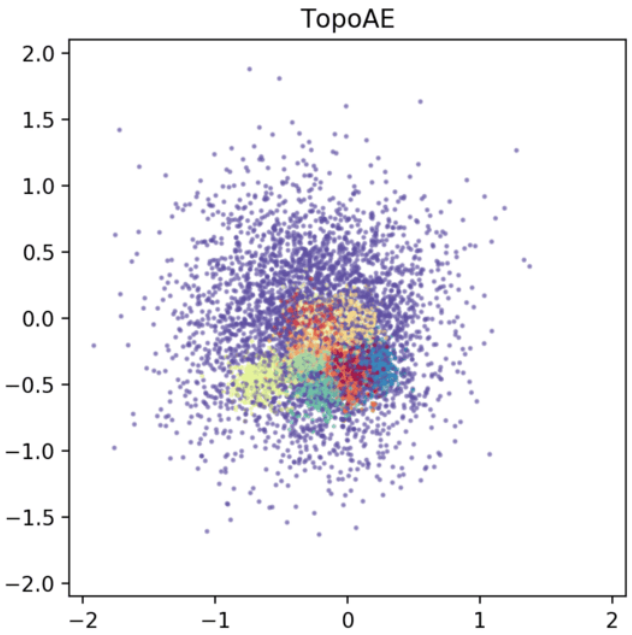

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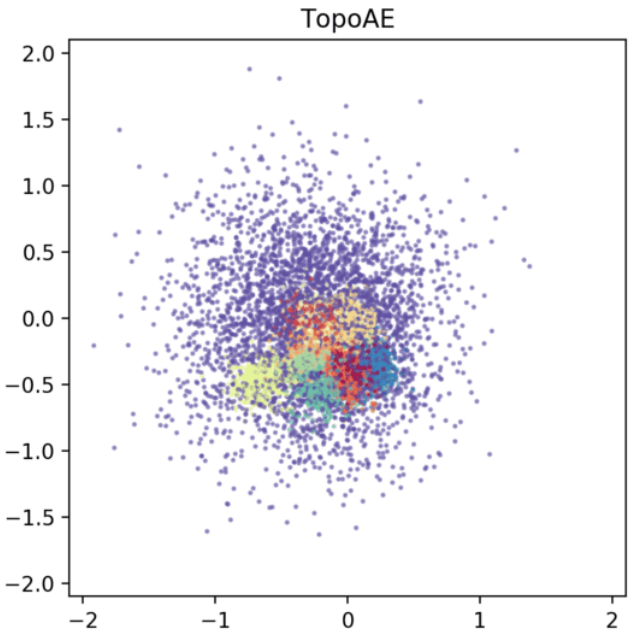

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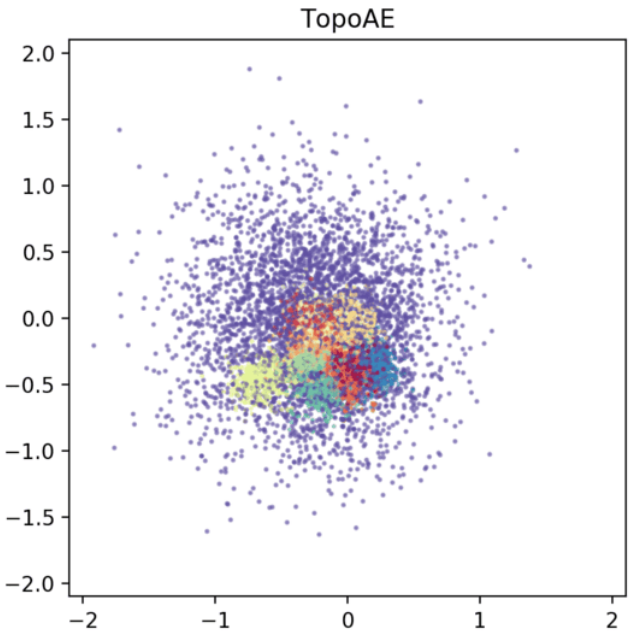

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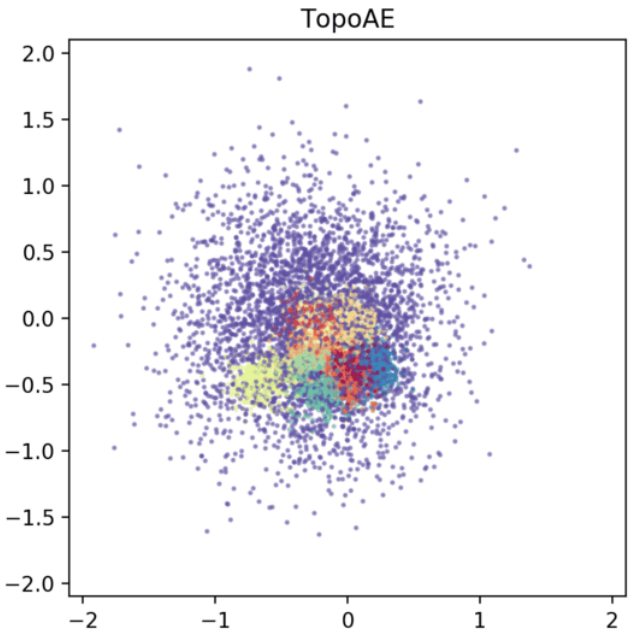

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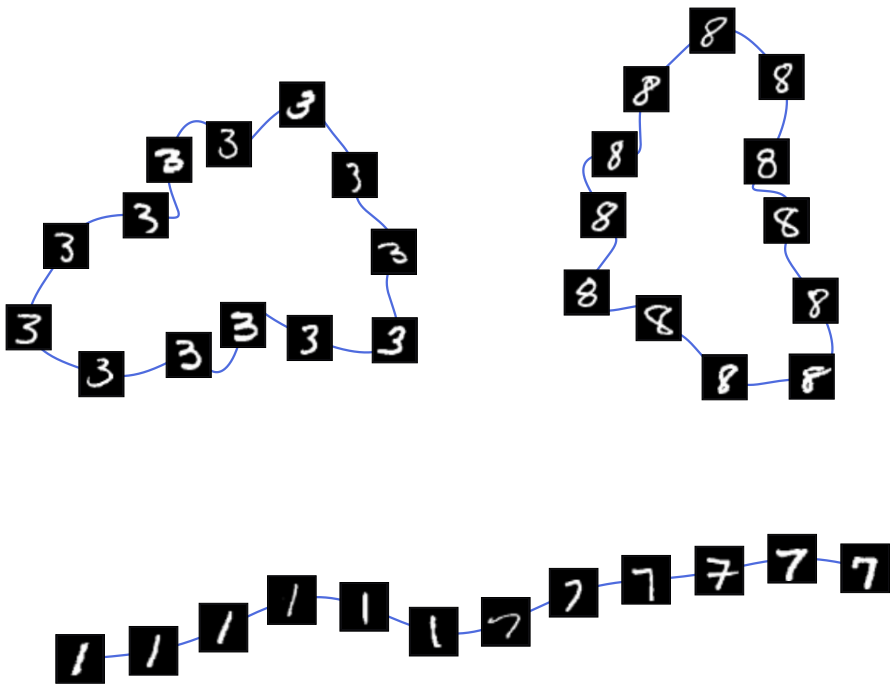
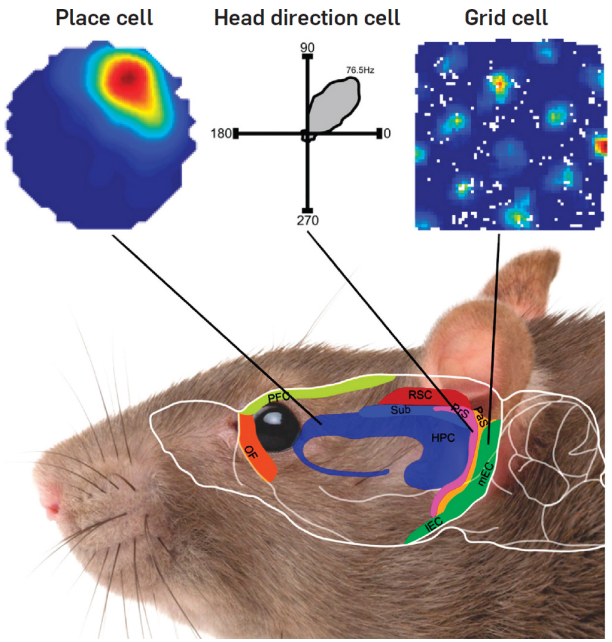
The Manifold Hypothesis

Claim: reasonable *real-world data* often lie concentrated around sub-manifolds of the ambient space.

- Example: mouse with S^1 freedom [GRG+21]
- Overly optimistic pseudo-example: MNIST





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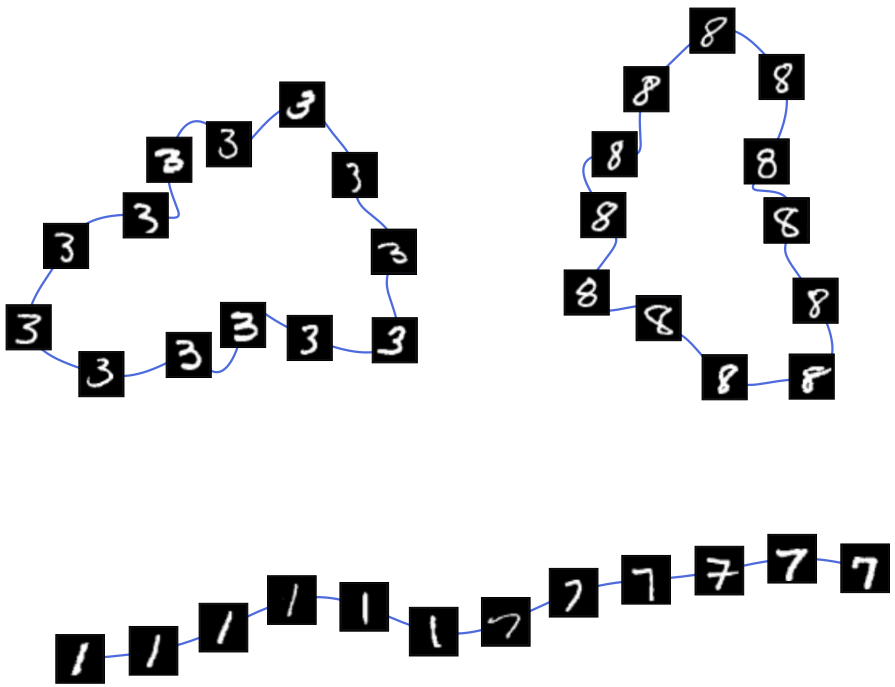

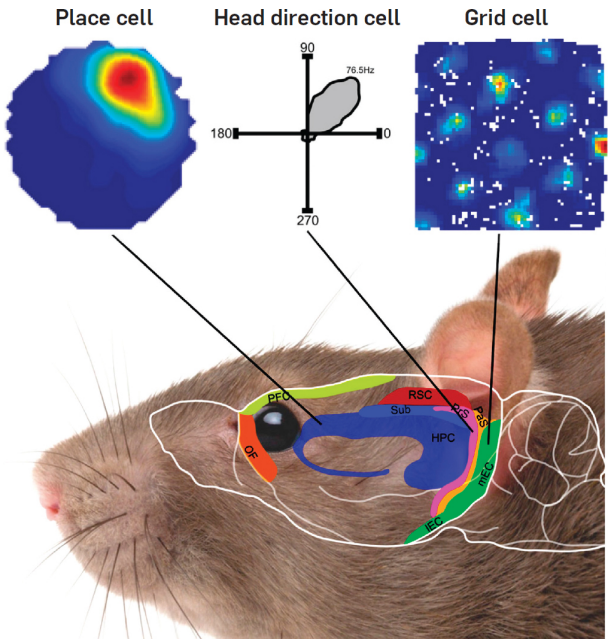

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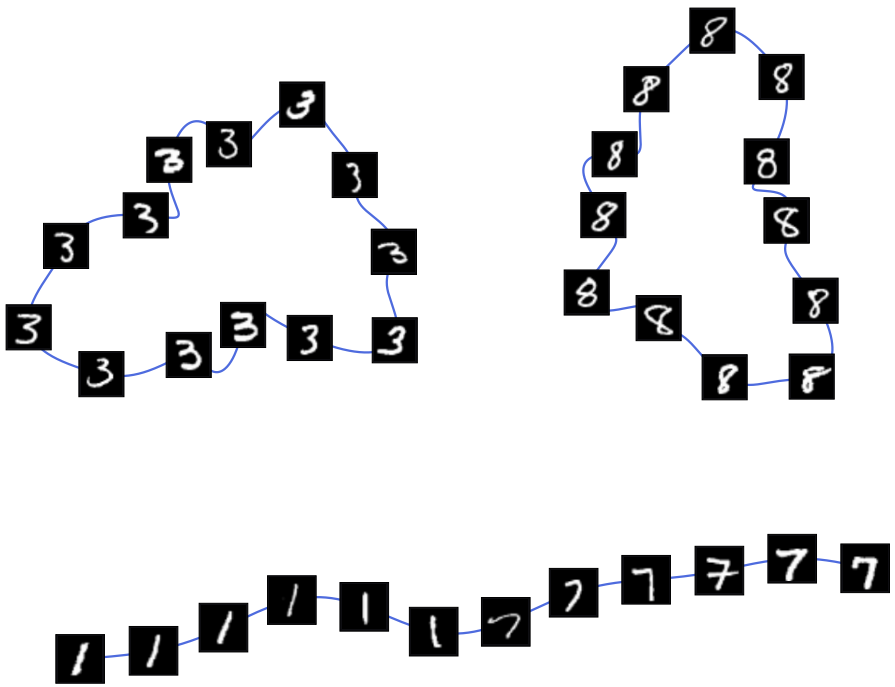

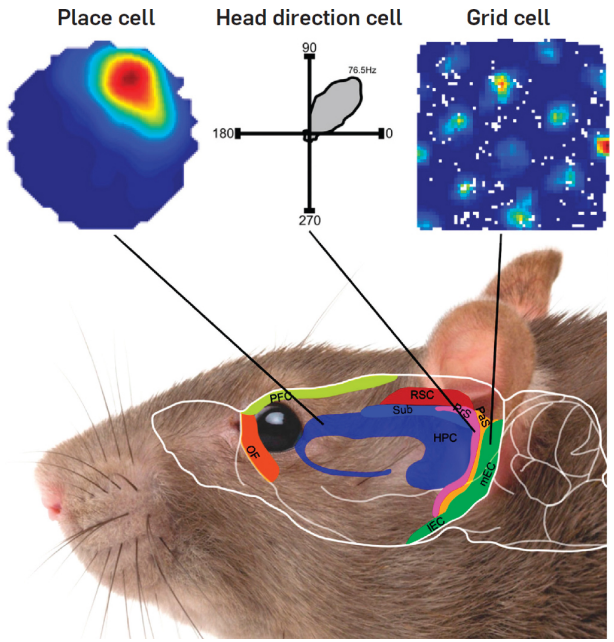

The Stratified* Manifold Hypothesis

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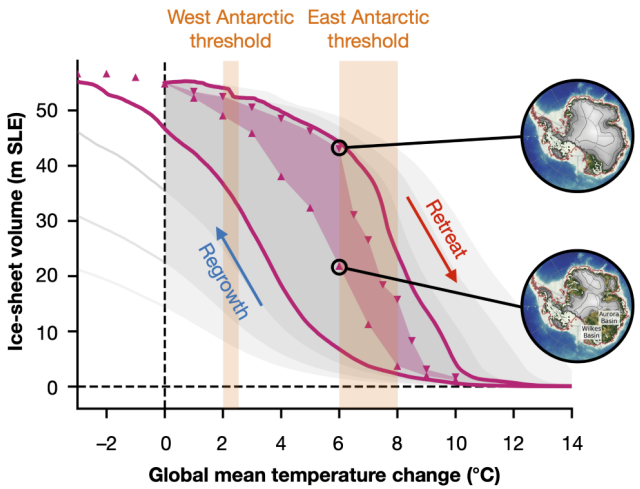
Stratified setting (with branching points et cetera) – more realistic.



Equation for ice cap thickness x_m:

$$\dot{x}_m = \frac{x_m^{1/2}}{H_0^{1/2}} \left[a_0 + \frac{2}{3}\lambda H_0^{1/2} x_m^{1/2} - \frac{1}{2}\mu x_m \right]$$

SS $\Longrightarrow$ bifurcation $\Longrightarrow$ hysteresis

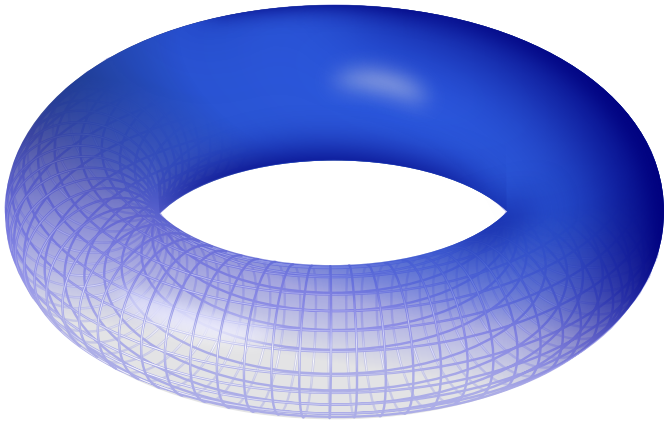
For the purposes of this talk: manifold-structured data.

* This is not widely accepted terminology. Yet.

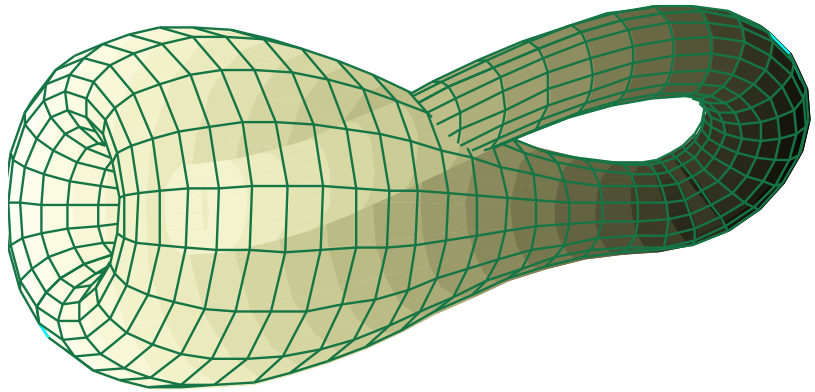
Definition (embedding)

A continuous map $f : X \to Y$ is said to be an *embedding* if it injective and a homeomorphism onto its image.

Example: $T^2 \hookrightarrow \mathbb{R}^3$



Non-example: $K \not\hookrightarrow \mathbb{R}^3$



Topological Issues

- Any d-manifold M with non-trivial homology in degree d is non-embeddable into $\mathbb{R}^d$. In particular:

Observation

Any closed orientable d-manifold is non-embeddable into $\mathbb{R}^d$.

Proof sketch:

- Without loss of generality M has one connected component.
- Therefore $H_0(M;\mathbb{R}) = \mathbb{R}$, and Poincaré duality yields

$$\mathbb{R} = H_0(M;\mathbb{R}) \cong H^d(M;\mathbb{R}) \cong H_d(M;\mathbb{R}),$$

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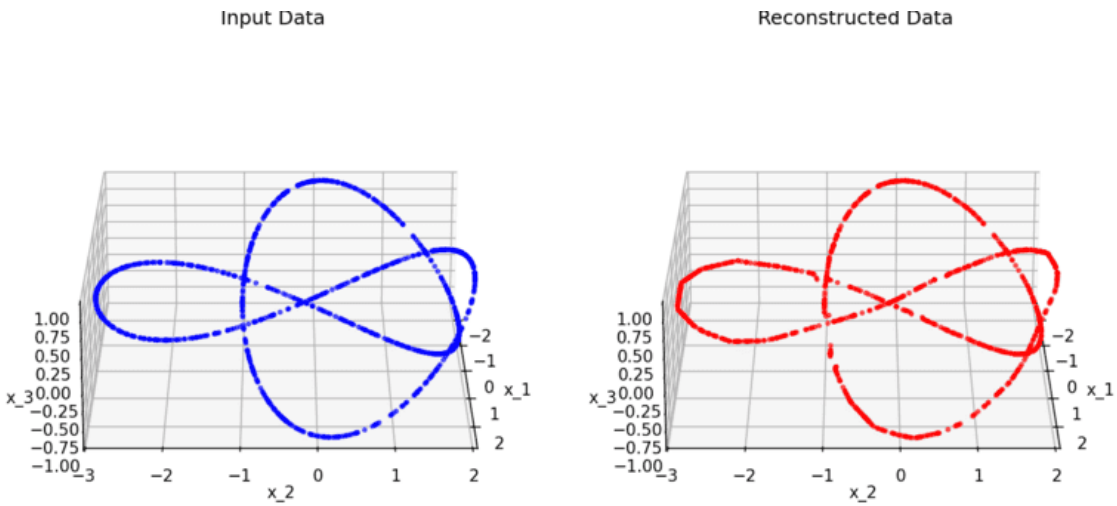
Our setting: *finite sample from a manifold*, not the manifold itself.

- $\exists$ a continuous function factoring through the latent space with perfect reconstruction in the *point cloud setting*
- UAT $\implies$ we can approximate this arbitrarily well



Takeaway: we *can* get the MSE loss arbitrarily small, but there will be areas of poor reconstruction and high bi-Lipschitz numbers.

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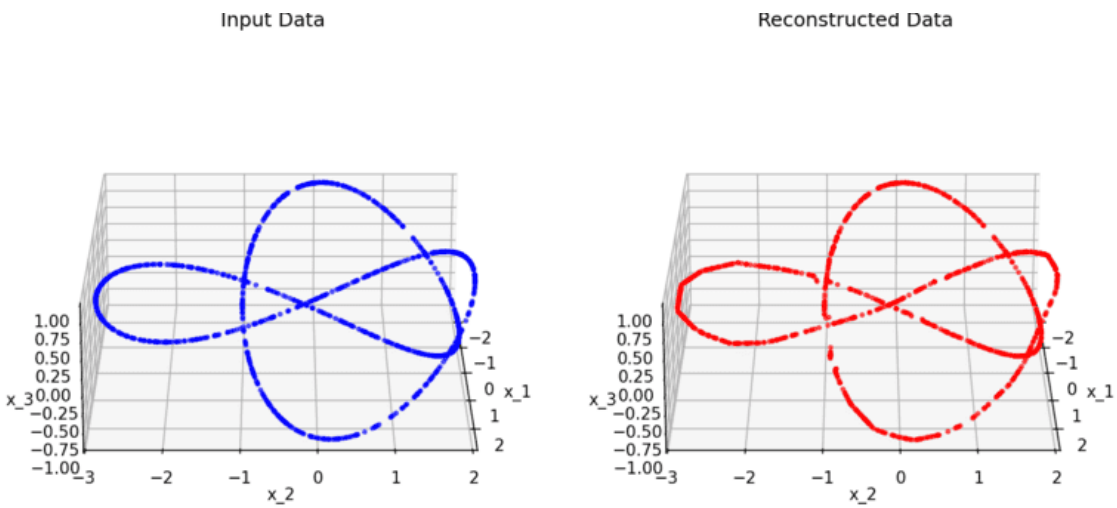

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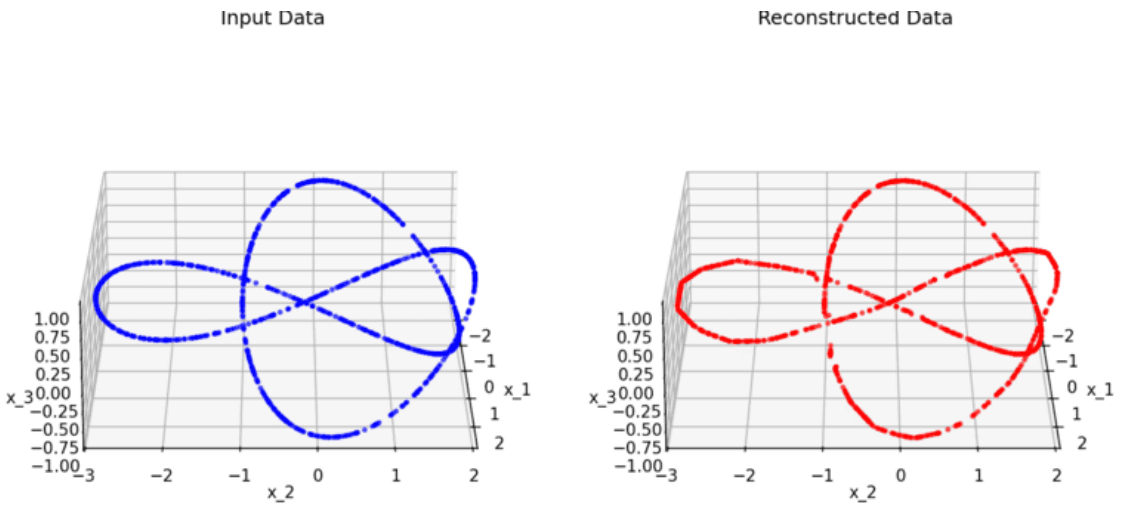

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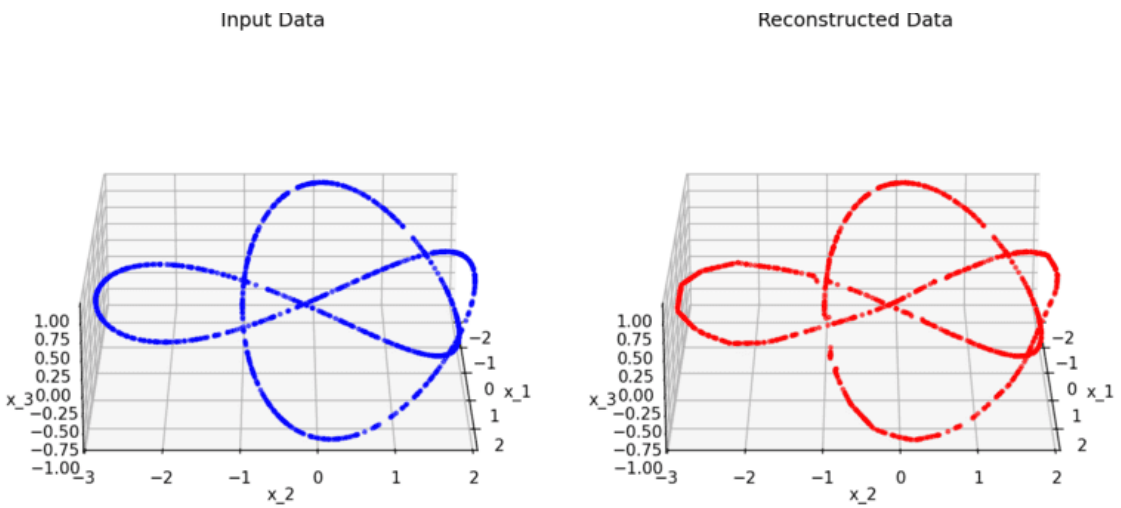

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The Solution: ManifoldAE

Instead of using one global latent space, use local latent spaces

- Learn patches that *are* embeddable
- Learn how these patches fit together

ManifoldAE – Mixture of Expert Autoencoders for Manifold Structured Data

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Joint work with Matthias Söls and Ulrich Bauer. Under review.

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Atlas of a Manifold

For a manifold M, and *atlas* is a collection of *charts*,

$$\mathcal{A} = \{(U_i, \varphi_i) \mid i \in I\},$$

where $\bigcup_{i \in I} U_i = M$, and $\varphi_i : U_i \to \varphi_i(U_i)$ is a homeomorphism.

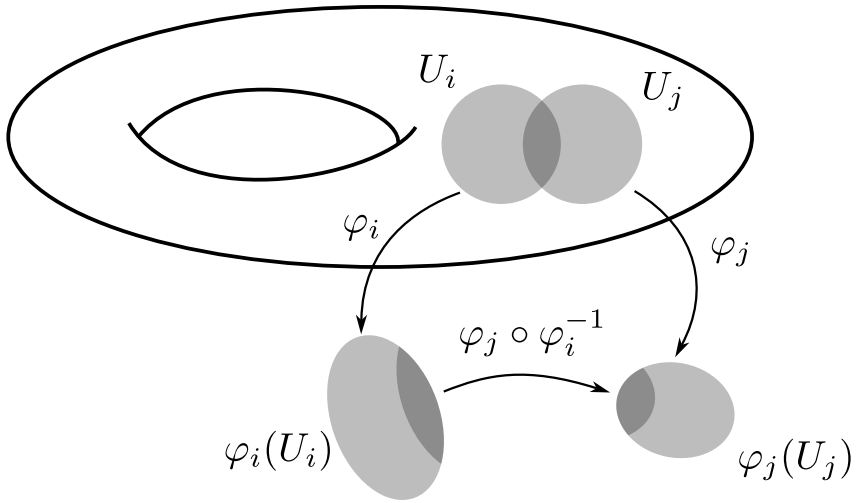


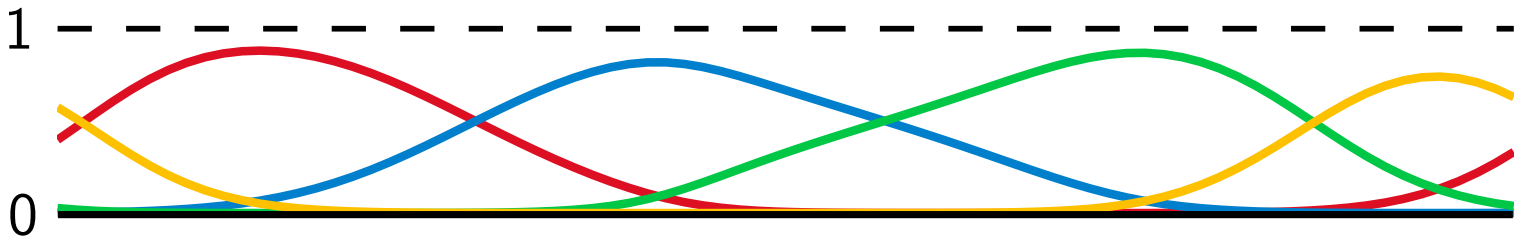
Figure: 2 charts of a torus atlas, and their corresponding *transition map*.

Partition of Unity

Definition (PoU)

A *partition of unity* of a topological space X is a set R of continuous maps $X \to [0,1]$ such that for every $x \in X$:

- there is a neighbourhood on which co-finitely many of the functions identically zero, and
- $\sum_{\rho \in R} \rho(x) = 1$ (the function values sum to one).



Finite case: this is a map $\omega : X \to \Delta^{n-1}$, whenever $|R| = n$.

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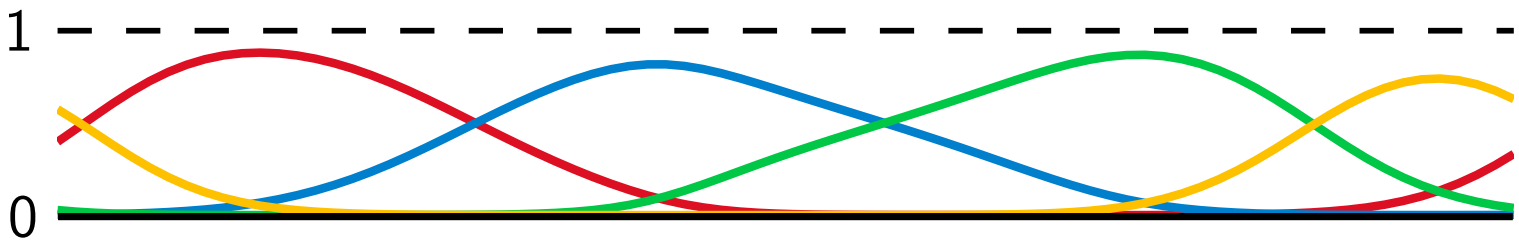

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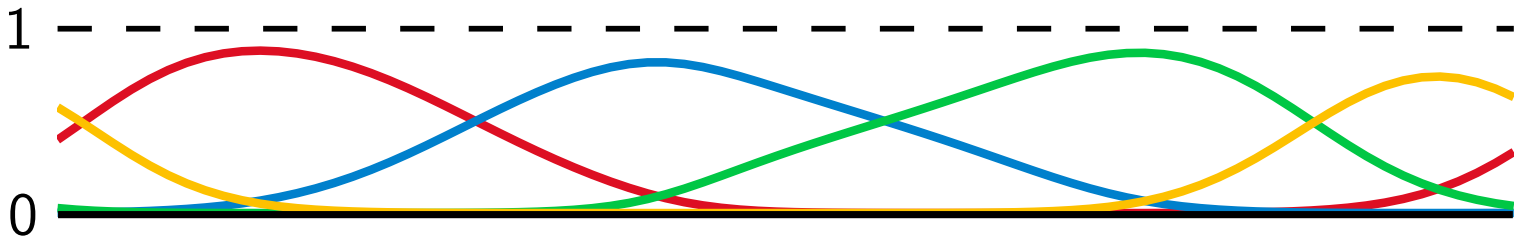

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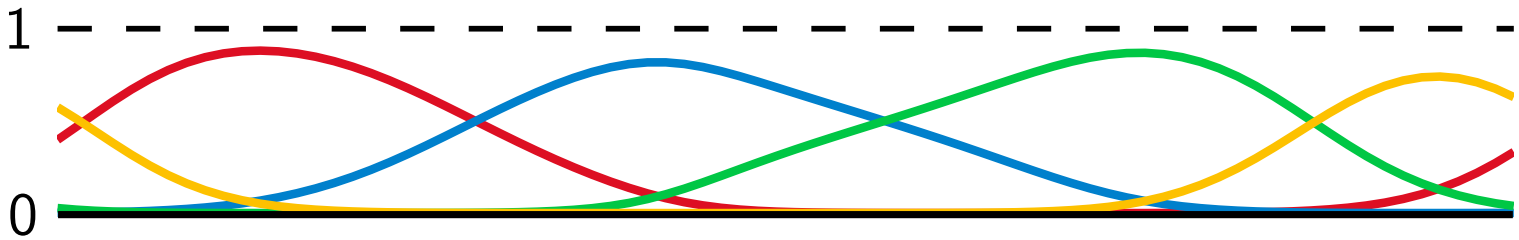

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Partition of Unity as a Gating Network

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Observation

Let $|I| < \infty$ and $\{\rho_i\}_{i \in I}$ be a PoU of X. Picking a sufficiently small threshold τ and letting $U_i = \{x \in X | \rho_i(x) > \tau\}$, we get an open cover of X.

Idea of ManifoldAE:

- Learn a PoU to be used as a gating network in a MoE model
- The AEs are local experts (on their respective cover element)
- Compositions $D_i \circ E_j \iff$ transition maps (how to glue)

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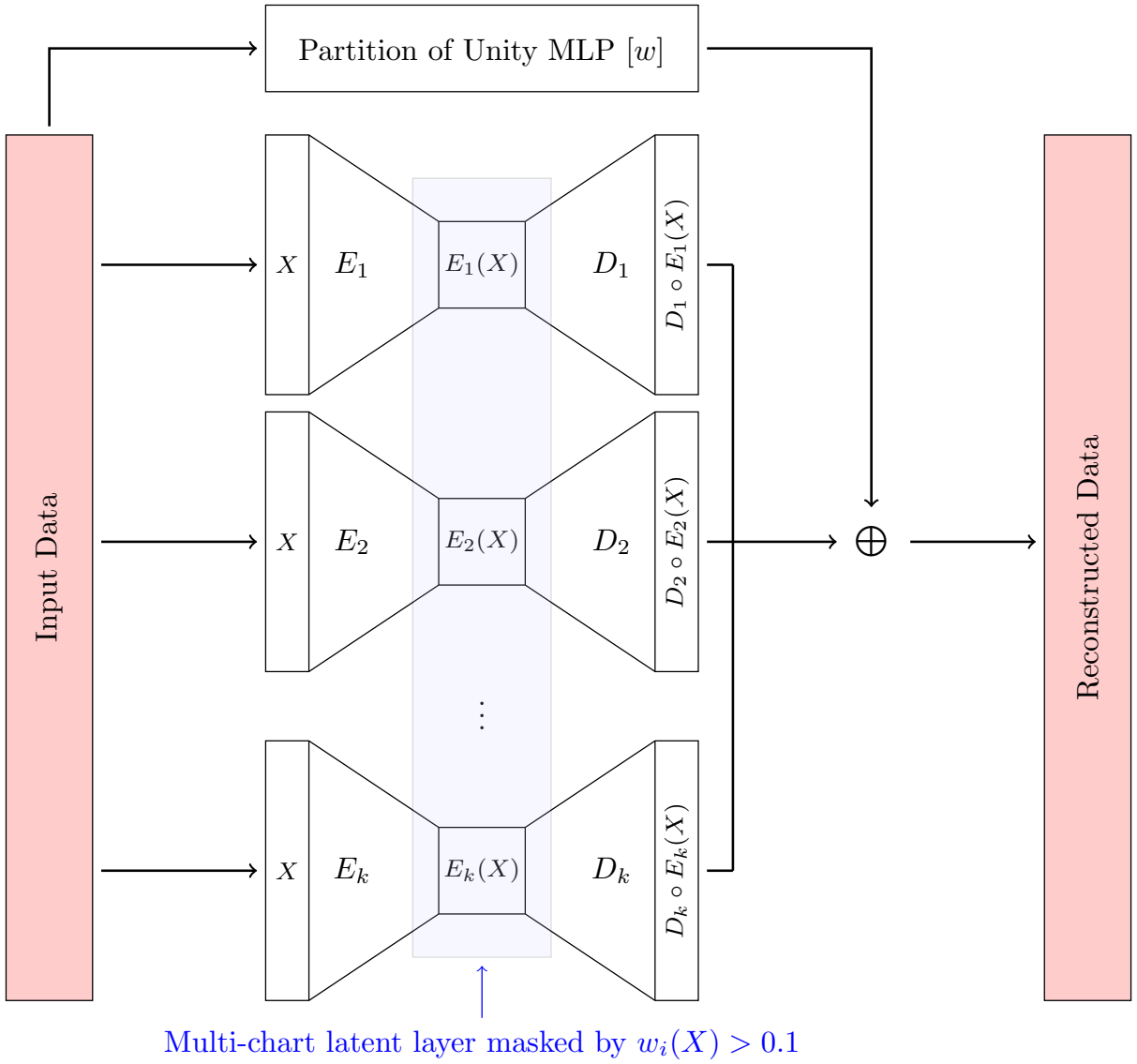
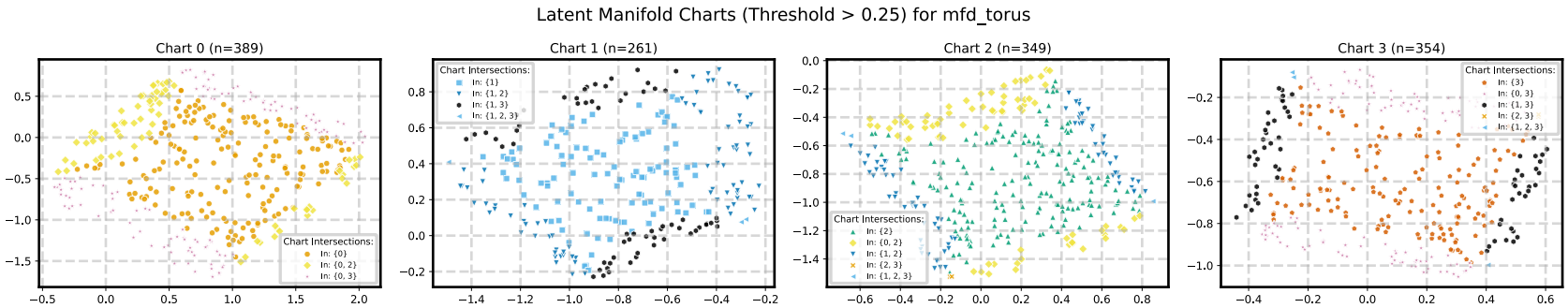
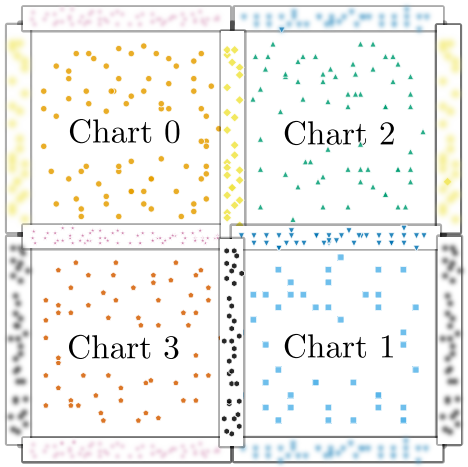
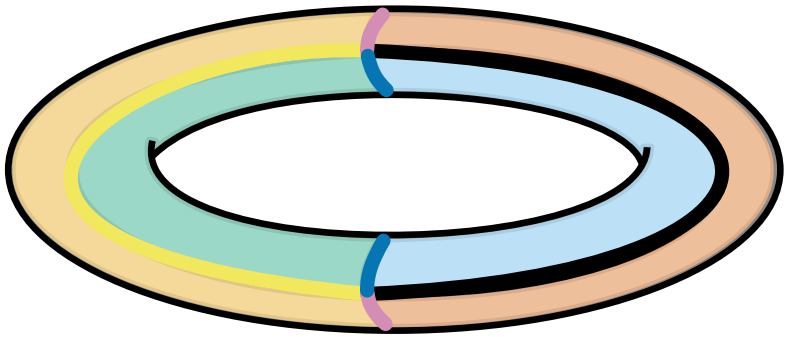
Model Architecture



Example Output of ManifoldAE







Overview of Loss Functions

Setting: finite sample $X \sim M \subseteq \mathbb{R}^N$, k autoencoders, ω_i is the i-th value of the PoU gate $\omega : X \to \Delta^{k-1}$.

We have the *weighted reconstruction loss:*

$$\mathcal{L} = \frac{1}{|X|} \sum_{x \in X} \sum_{i=1}^{k} \omega_i(x) \|x - D_i \circ E_i(x)\|^2.$$

Additional loss terms:

- bi-Lipschitz loss
- Conformal loss
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bi-Lipschitz loss

Want: discourage "ripping" and "squashing". Letting $f_i := D_i \circ E_i$, the quotient

$$Q_i(x, y) := \frac{||f_i(x) - f_i(y)||}{||x - y||}$$

should, in other words, be as close to 1 as possible (on average).

Loss function to this end:

$$\mathcal{L}_{\text{lip}} = \frac{1}{|\text{NNG}|} \sum_{i=1}^{k} \sum_{(x,y)\in \text{NNG}} \omega_i(x)\,\omega_i(y) \underbrace{\log\left(\frac{Q_i(x,y)}{2} + \frac{1}{2Q_i(x,y)}\right)}_{\text{minimum at } Q_i(x,y)=1}$$

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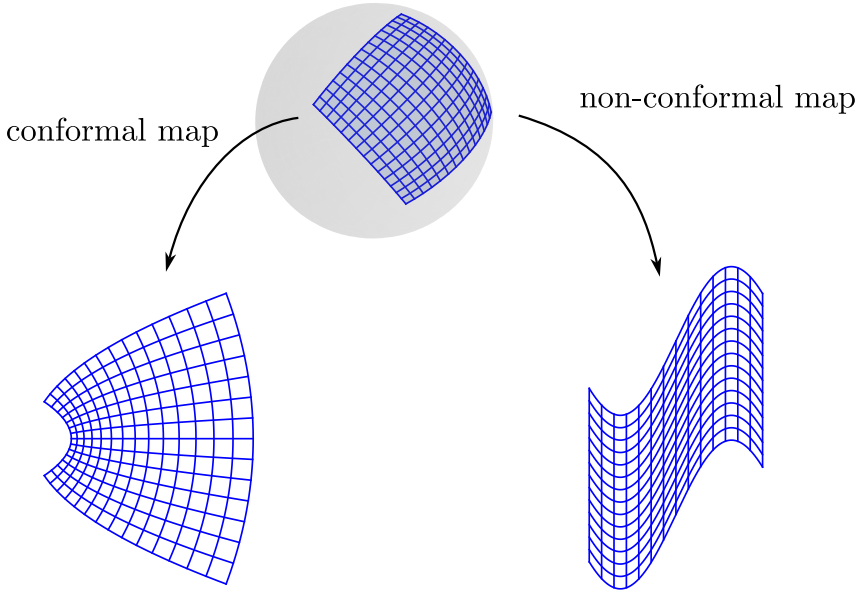
Conformal loss

Encourage *geometric regularity*: locally even stretching in latents



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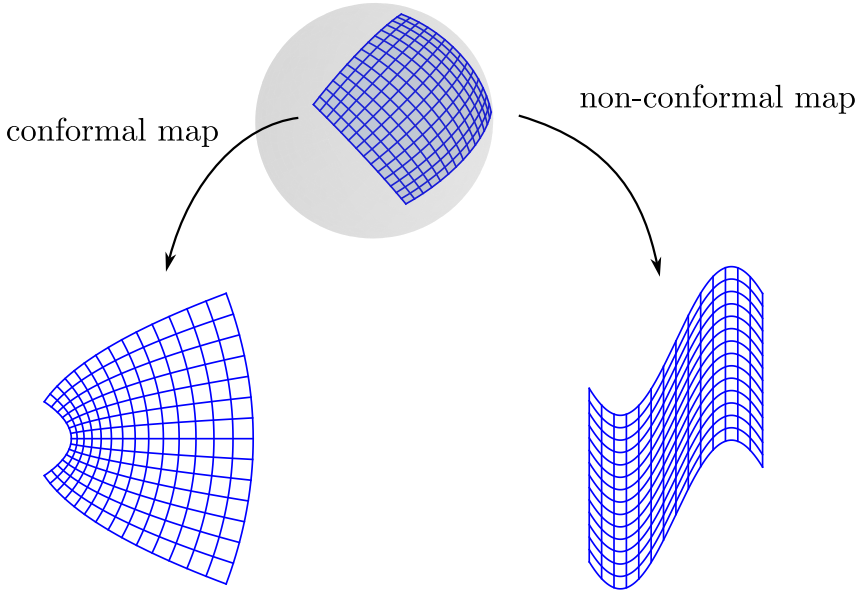

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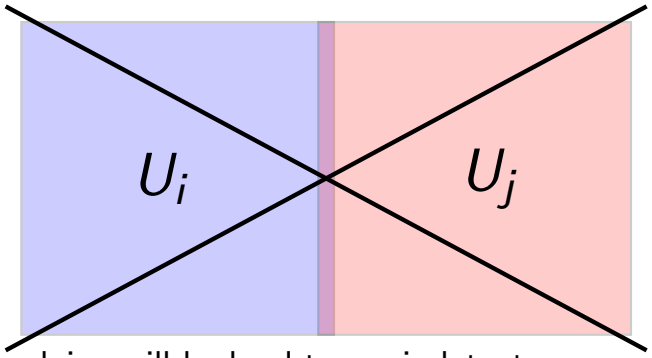
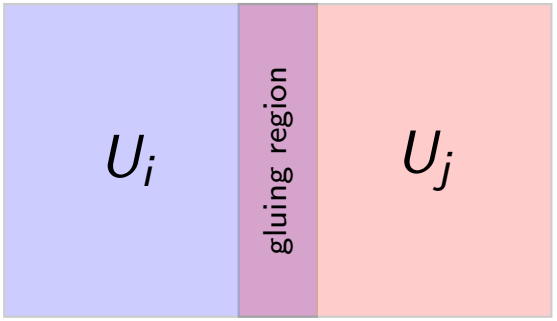
Smooth partition loss



gluing will be hard to see in latent spaces



sufficiently large gluing regions

Dirichlet energy of the PoU (c.f. Heat equation):

$$E[\omega] = \frac{1}{2} \int_M \|\nabla \omega(x)\|^2 \, dx.$$

Approximation will serve as our loss function:

$$\mathcal{L}_{\text{partition}} = \frac{1}{|\text{NNG}|} \sum_{(x,y)\in \text{NNG}} \sum_{i=1}^{k} \left(\frac{w_i(x) - w_i(y)}{\|x - y\|} \right)^2 .$$

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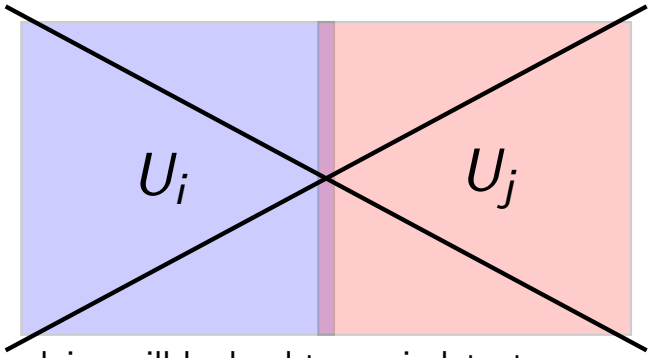

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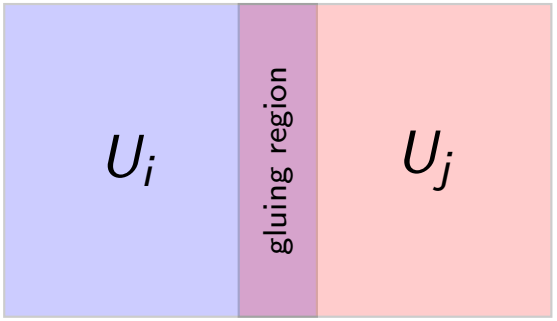

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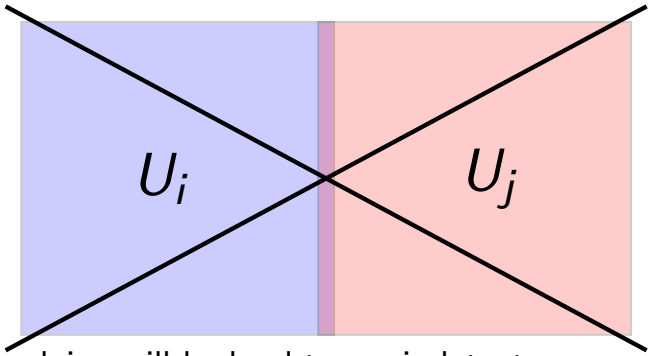

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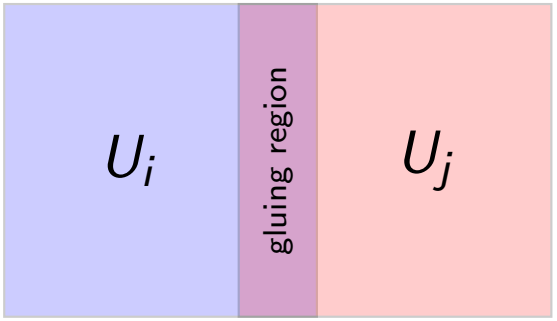

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Training procedure

Munich Center for Machine Learning

1. Pre-training of PoU gating network
   - Pick target distribution via FPS and GMM (using graph distances).
   - Minimise KL divergence over 500 epochs.
2. Pre-training to prevent mode collapse
   - Freeze PoU gate and train the AEs for 2000 epochs.
   - No need to compute $\mathcal{L}_{\text{partition}}$ here.
3. Final training cycle
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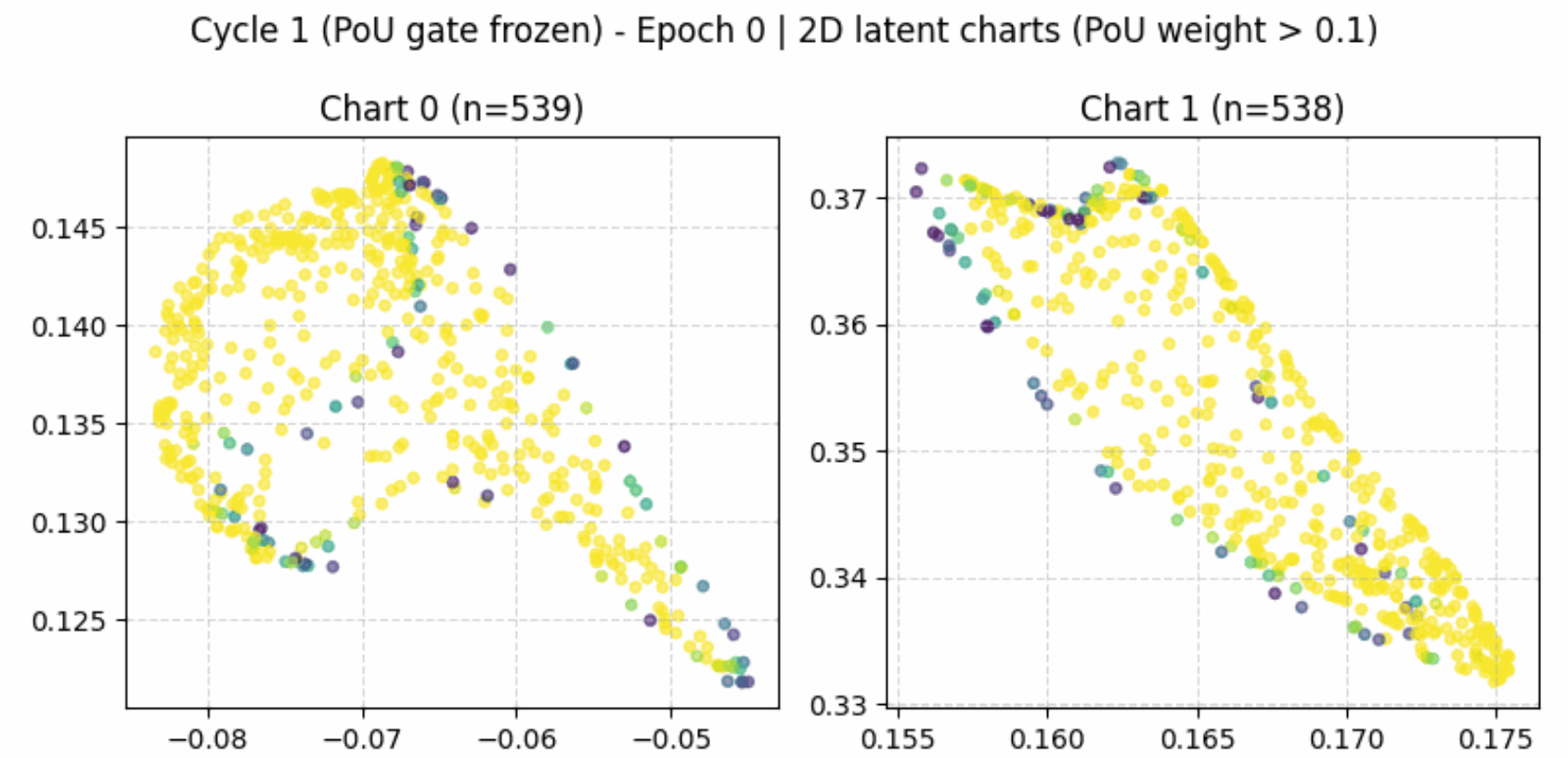

Measuring performance

- Using max SE loss and bi-Lipschitz estimate for evaluation, we outperform all single-chart methods (by ~ 2 OOM).
- Non-trivial to do standard NLDR metrics (*continuity*, *trustworthiness* etc.) because they are designed for single-latent space setups.

Fix:

- Consider test dataset X containing 200 points.
- For each subset $X_i = \{x \in X \mid w_i(x) > 0.1\}$ with ≥ 25 points, compute the given metric on $E_i \mid_{X_i} : X_i \to \mathbb{R}^d$.
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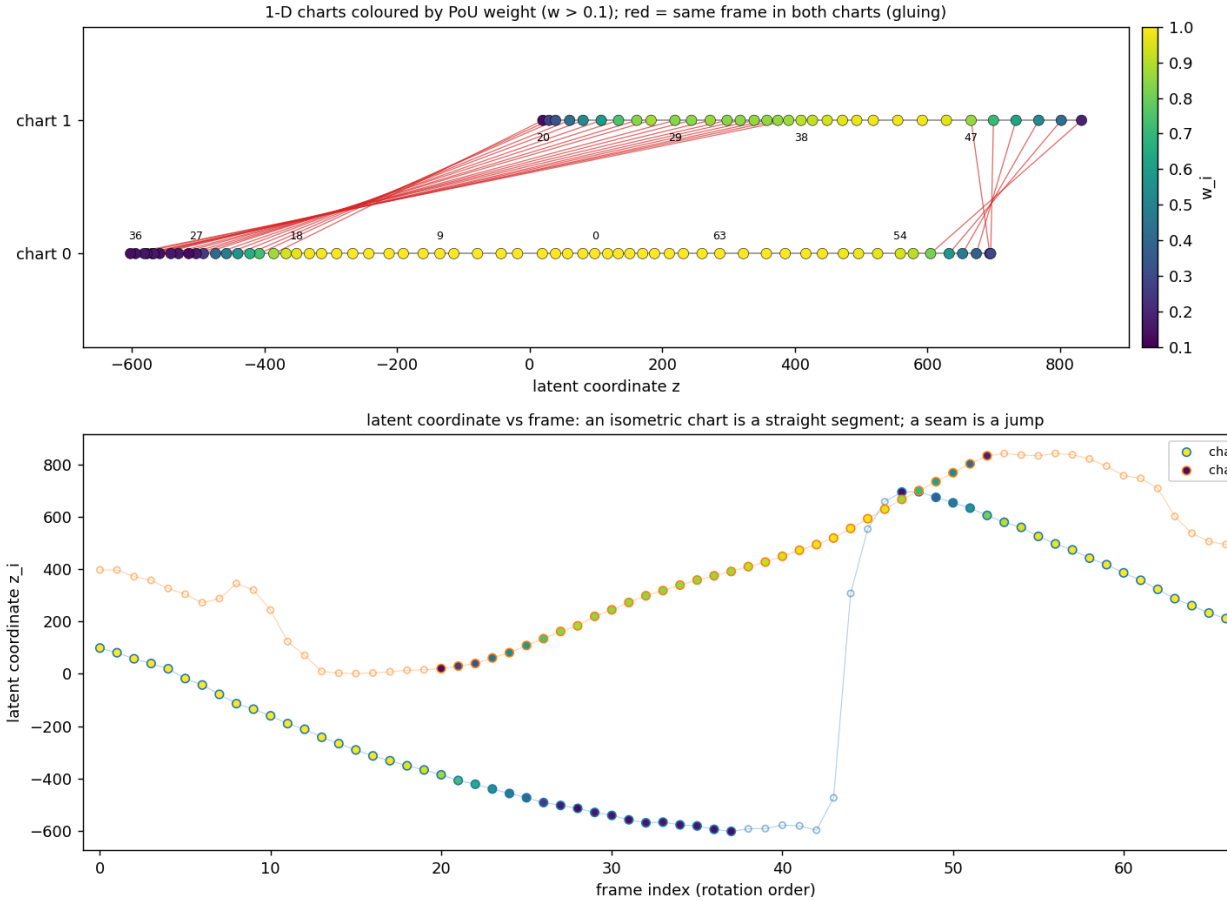
High-dimensional real-world data



COIL-20 Rubber Duck

(Ground Truth: S^1)

Loose ends

- More experiments on real-world datasets - how reliable is this?
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Thanks for your attention!

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Happy to take questions! Preprint coming soon (I hope).

References

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